POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

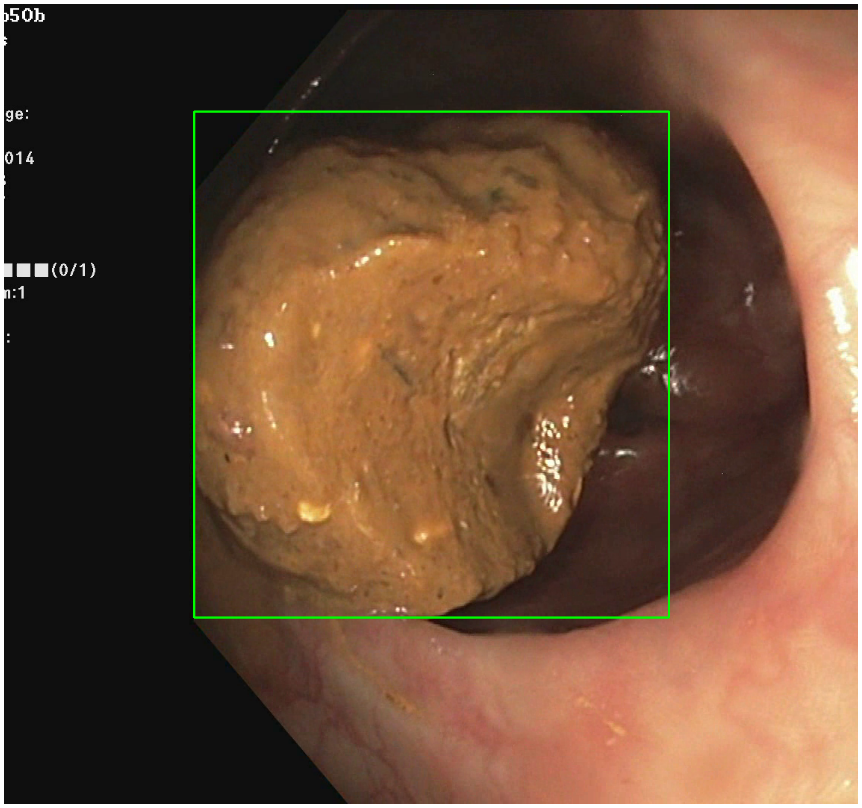
Video ID:	3f450a06-3397-48ed-ab27-5bd184af2862
Total Frames:	40
Frame Rate:	25.0 fps (assumed)
Duration:	1.6 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	20	✓
RT-DETR Detections	23	✓
MedSAM2 Detections	23	✓
Consensus (3-Model Agreement)	1	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	1	100.0%
TOTAL ANALYZED	1	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 20 frames)



Feature	Value	Interpretation
Frame Index	24	Frame 24 of 40
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 73.6%
Redness Score	0.144	Color-based vascularization
Relative Radius	35.8%	Polyp size as % of frame diagonal
Texture Score	0.043	Surface morphology
Vessel Visibility	0.005	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket

Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	93.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:
A total of 1 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 1 (100.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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